Test and Truck Data Signal Processing

Sesha Charla

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Contents

1 Recursive Least Squares with Change Detection [2]

The goal of the present work is to remove the high-frequency measurement noise from the test and truck measurement signals of concentrations, temperature, flow-rate and urea injection rate. This signals while slowly varying, undergo abrupt changes due to the nonlinearity of the system dynamics. Thus, the problem becomes rejecting the high-frequency noise while retaining the signal jumps. This is achieved by using a recursive least-squares filter with forgetting factor which is set to zero for one sample based on the detection of the jump in magnitude. The Cumulative Sum (CUMSUM) algorithm, a special case of Sequential Probability Ratio Test (SPRT) is used for detecting the signal jumps. The algorithm and the tuning method is described in the rest of this section. Let θ be the signal to be estimated, y be the noisy measurement signal, ε be the noise. We have the signal model:

$$
\begin{aligned}
y_t &= \theta_t + \varepsilon_t \\
\hat{\theta}_t &= \lambda\hat{\theta}_{t-1} + (1-\lambda)y_t \\
\epsilon_t &= y_t - \hat{\theta}_{t-1} \\
s_t^{(1)} &= \epsilon_t \\
s_2^{(2)} &= -\epsilon_t \\
g_t^{(1)} &= \max\left(g_{t-1}^{(1)} + s_t^{(1)} - \nu, 0\right) \\
g_t^{(2)} &= \max\left(g_{t-1}^{(2)} + s_t^{(2)} - \nu, 0\right)
\end{aligned}
$$

$$
\begin{aligned}
\text{Alarm if } g_t^{(1)} > h \text{ or } g_t^{(2)} > h \\
\text{After alarm, } g_t^{(1)} = 0,\ g_t^{(2)} = 0 \text{ and } \hat{\theta}_t = y_t
\end{aligned}
$$

$$
\begin{aligned}
\textbf{\textit{Design Parameters}} &: \lambda, \nu, h \\
\textbf{\textit{Output}} &: \hat{\theta}_t
\end{aligned}
$$

Algorithm 1: Recursive Least Squares with Change Detection

1.1 Tuning of CUMSUM-RLS filtering algorithm [2]

Start with a very large threshold h. Choose ν to be one half of the expected change, or adjust ν such that $g_t = 0$ more than 50% of the time. Then set the threshold so the required number of false alarms (this can be done automatically) or delay for detection is obtained.

- If faster detection is sought, try to decrease ν.
- If fewer false alarms are wanted, try to increase ν.
- If there is a subset of the change times that does not make sense, try to increase ν.

1.2 Equivalent time constant approximation for forgetting factor [4]

The forgetting factor in RLS makes it behave like a low-pass filter. The equivalent time constant for the forgetting factor is given by:

$$
\begin{aligned}
&e^{-t_d/T_f} = \lambda \\
\implies &T_f = \frac{-t_d}{\ln(\lambda)} \approx \frac{t_d}{(1-\lambda)} = \begin{cases} 100t_d & \text{for } \lambda = 0.99 \\ 10t_d & \text{for } \lambda = 0.9 \end{cases}
\end{aligned}
$$

where, t_d is the sampling period and T_f is the equivalent time constant for the forgetting factor. It is thus clear that with a forgetting factor of $\lambda = 0.99$ approximately 100 to 400 immediately previous samples are used by the current parameter estimation. If $\lambda = 0.9$ that number goes down to 10 to 40.

λ, ν and h are chosen such a way that the power spectral density (calculated using Welch's method [3], [1]) remains intact within the frequency range of interest. The ν and h values depend on the actual change magnitudes of the individual signals. These ranges are different for test and truck conditions. Thus, these values and λ are tuned independently for truck and test data sets.

Part I
Test Data Signal Processing

2 Power Spectral Density of Individual Test Signals

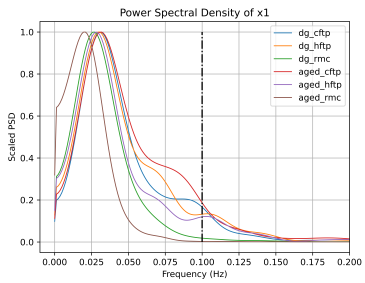

Figure 1: PSD of $[NO_x]^{out}$ FTIR signal

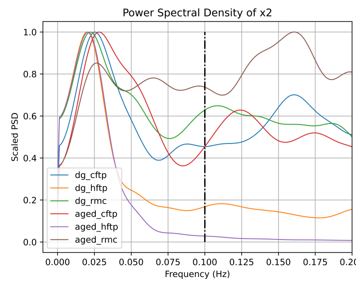

Figure 2: PSD of $[NH_3]^{out}$ FTIR signal

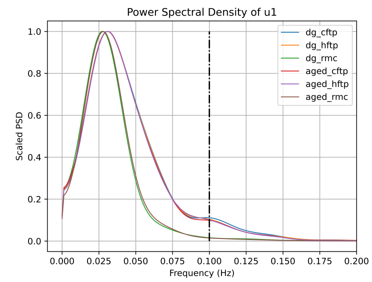

Figure 3: PSD of $[NO_x]^{in}$

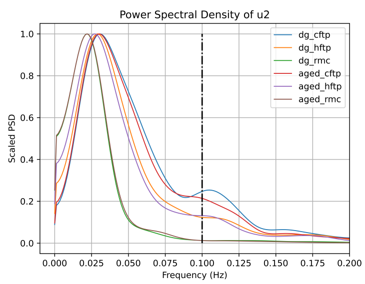

Figure 4: PSD of urea injection rate

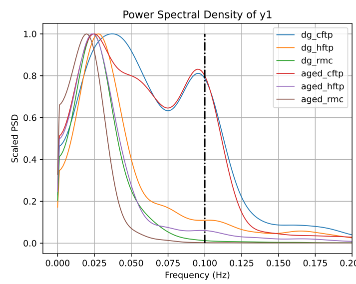

Figure 5: PSD of $[NO_x]^{out}$ measurement signal

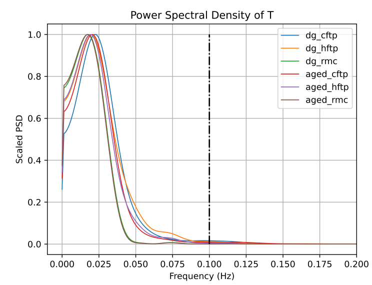

Figure 6: PSD of temperature

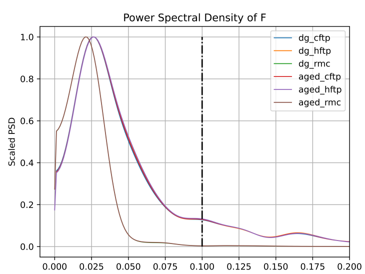

Figure 7: PSD of mass flow rate

2.1 Forgetting factor for RLS

The PSD plots show that much of the energy for the signals in within the frequency range of $0\,Hz$ to $0.1\,Hz$. Thus, the forgetting factor for the RLS algorithm is chosen such that the equivalent time-constant is greater than $1/1 = 10\,s$. We have,

$$
\begin{aligned}
T_f &= \frac{t_d}{1-\lambda} \\
T_f = 10 \quad &\text{and} \quad t_d = 0.2 \\
\implies \lambda &= 1 - \frac{t_d}{T_f} = 1 - \frac{0.2}{10} = 0.98
\end{aligned}
$$

2.2 h and ν parameters for individual signals

signal	h	ν
x_1	1	0.5
x_2	0.04	0.04
u_1	1	0.5
u_2	0.2	0.1
T	40	30
F	50	20
y_1	1	0.5

Table 1: Values of threshold (h) and bias (ν) for individual test data signals

3 Degreened CFTP

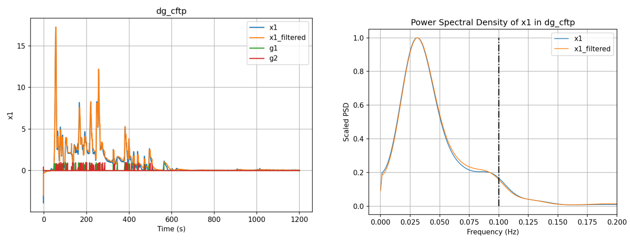

Figure 8: Filtered results for x_1

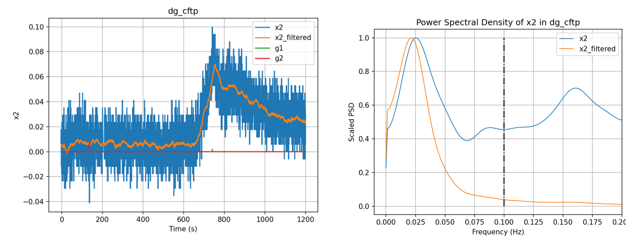

Figure 9: Filtered results for x_2

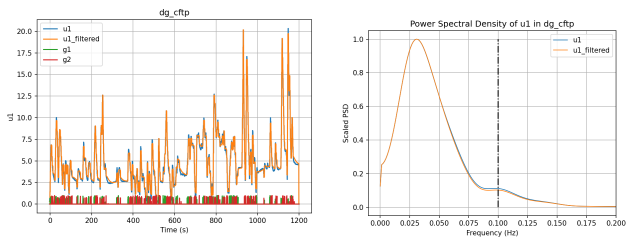

Figure 10: Filtered results for u_1

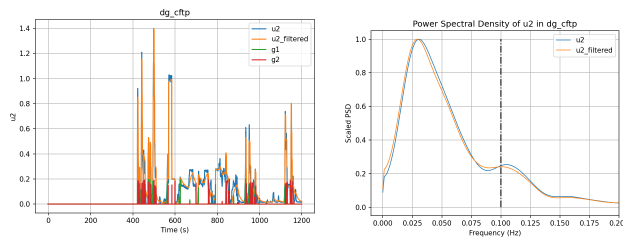

Figure 11: Filtered results for u_2

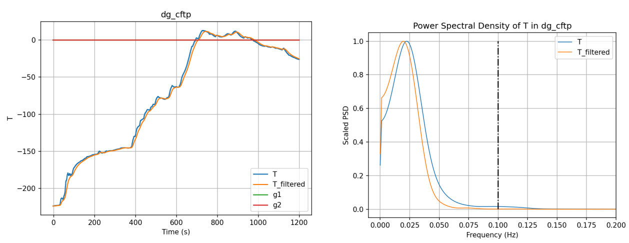

Figure 12: Filtered results for T

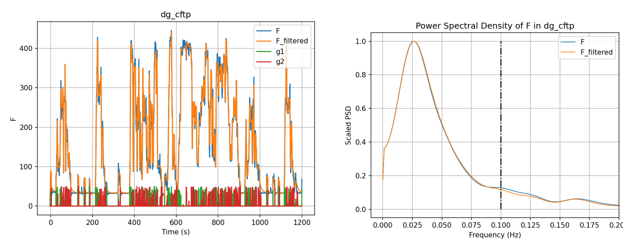

Figure 13: Filtered results for F

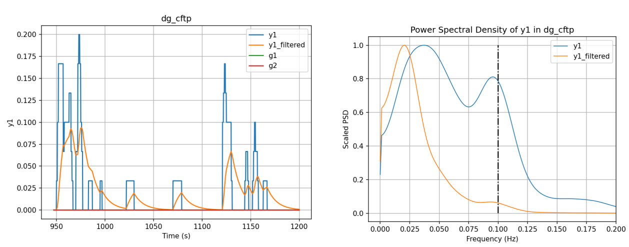

Figure 14: Filtered results for y_1

4 Degreened HFTP

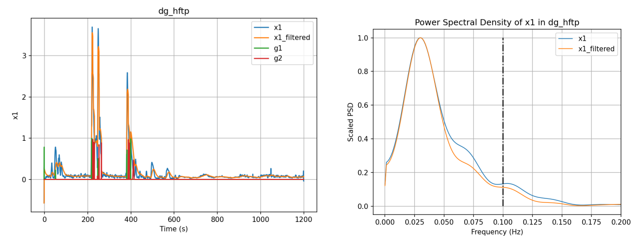

Figure 15: Filtered results for x_1

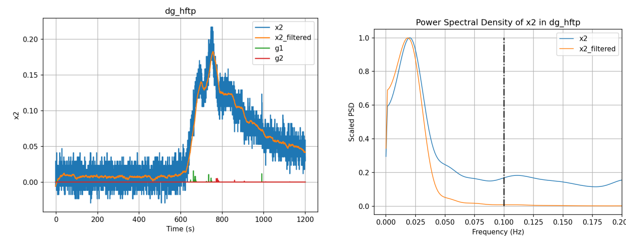

Figure 16: Filtered results for x_2

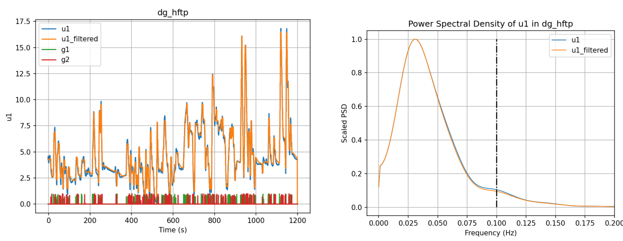

Figure 17: Filtered results for u_1

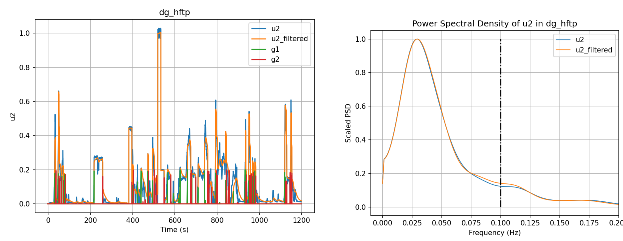

Figure 18: Filtered results for u_2

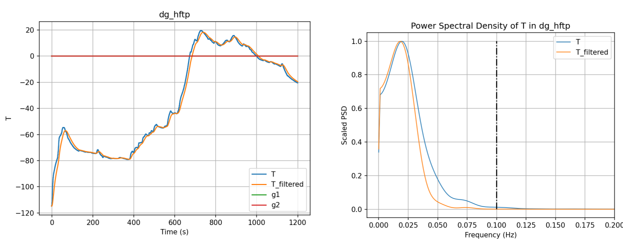

Figure 19: Filtered results for T

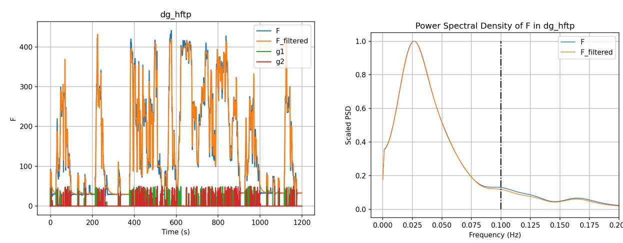

Figure 20: Filtered results for F

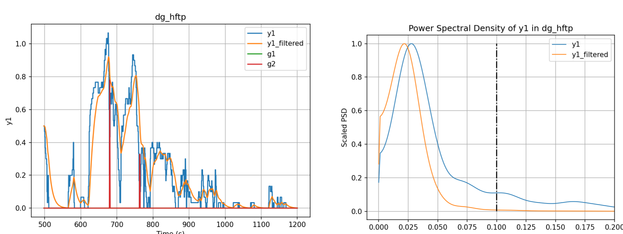

Figure 21: Filtered results for y_1

5 Degreened RMC

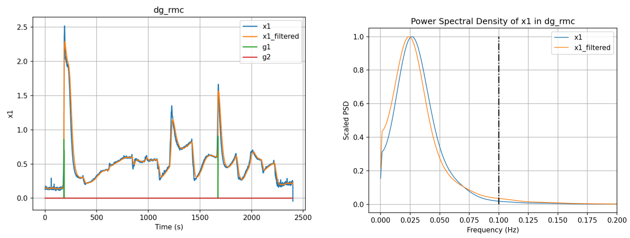

Figure 22: Filtered results for x_1

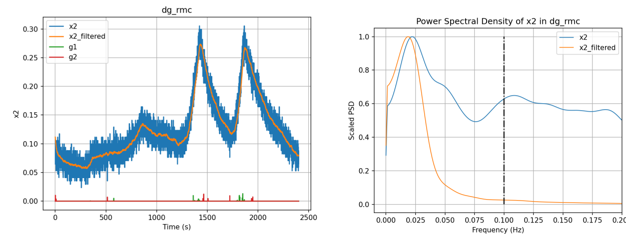

Figure 23: Filtered results for x_2

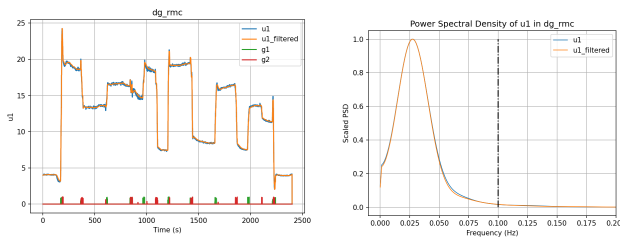

Figure 24: Filtered results for u_1

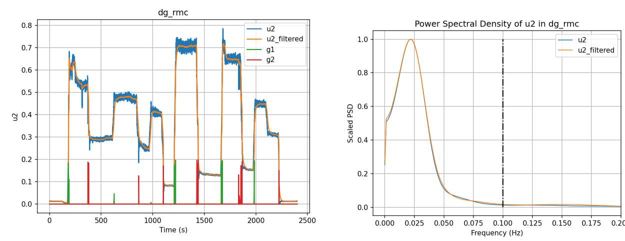

Figure 25: Filtered results for u_2

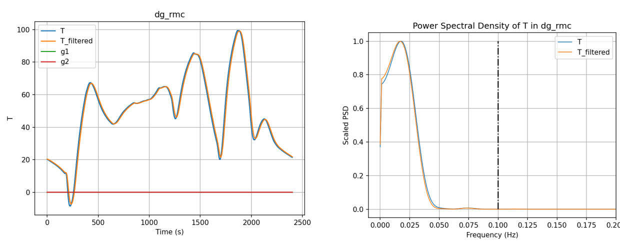

Figure 26: Filtered results for T

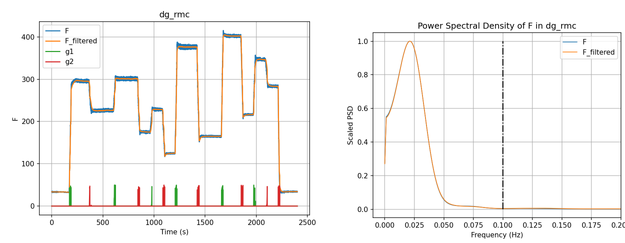

Figure 27: Filtered results for F

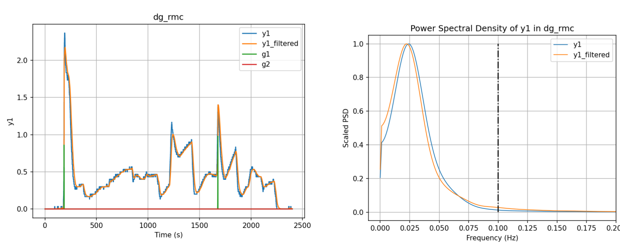

Figure 28: Filtered results for y_1

6 Aged CFTP

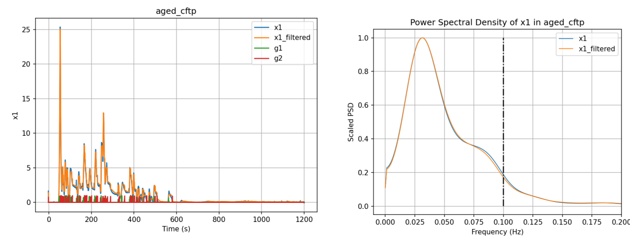

Figure 29: Filtered results for x_1

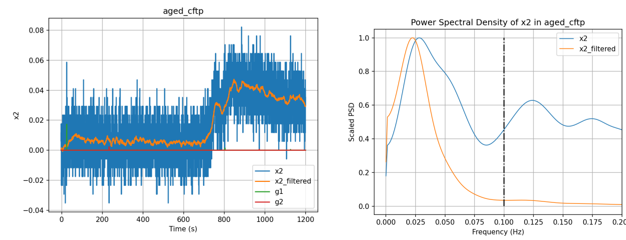

Figure 30: Filtered results for x_2

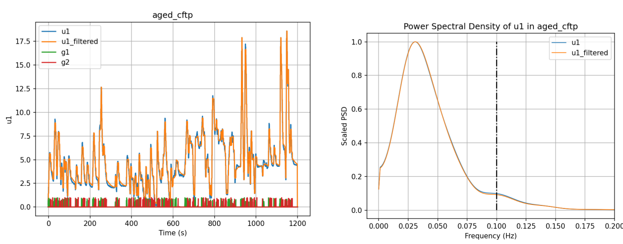

Figure 31: Filtered results for u_1

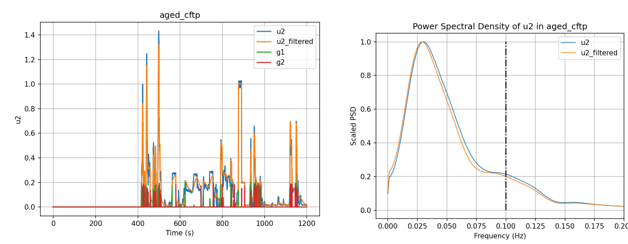

Figure 32: Filtered results for u_2

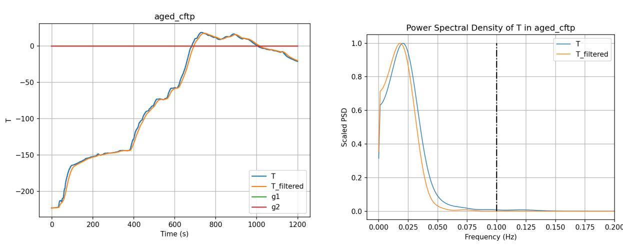

Figure 33: Filtered results for T

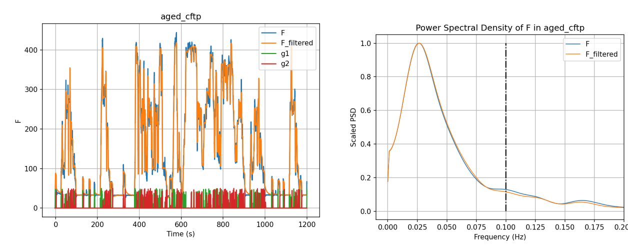

Figure 34: Filtered results for F

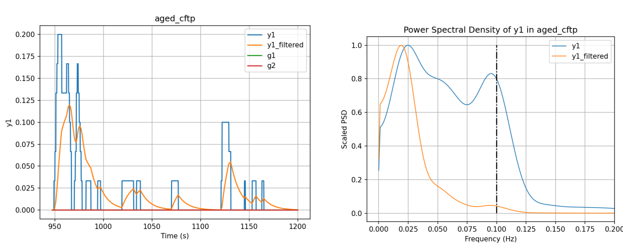

Figure 35: Filtered results for y_1

7 Aged HFTP

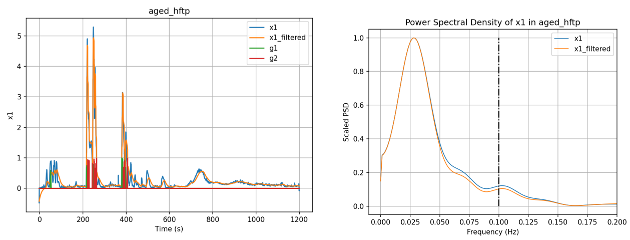

Figure 36: Filtered results for x_1

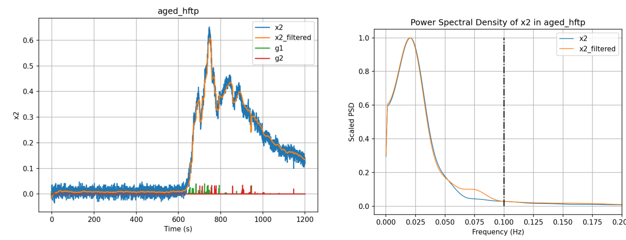

Figure 37: Filtered results for x_2

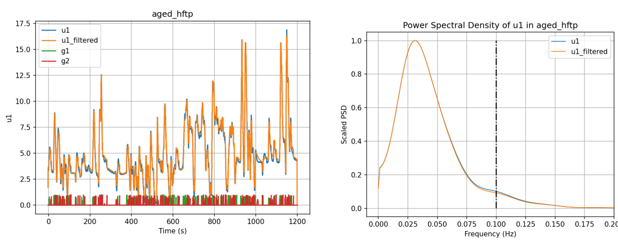

Figure 38: Filtered results for u_1

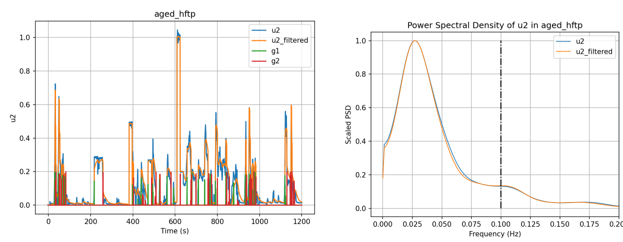

Figure 39: Filtered results for u_2

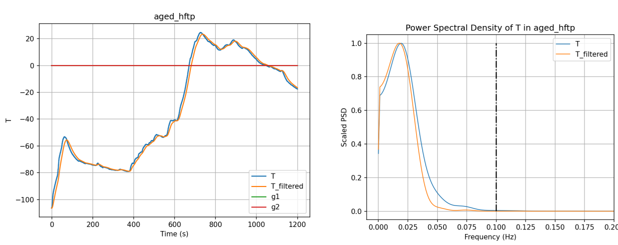

Figure 40: Filtered results for T

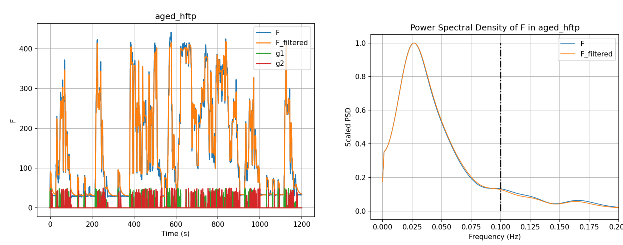

Figure 41: Filtered results for F

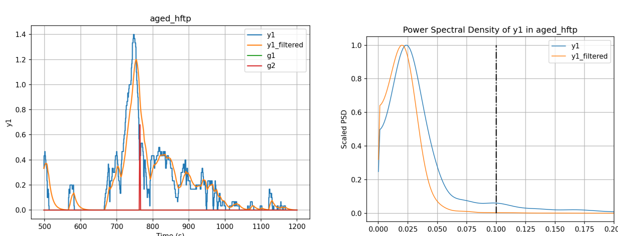

Figure 42: Filtered results for y_1

8 Aged RMC

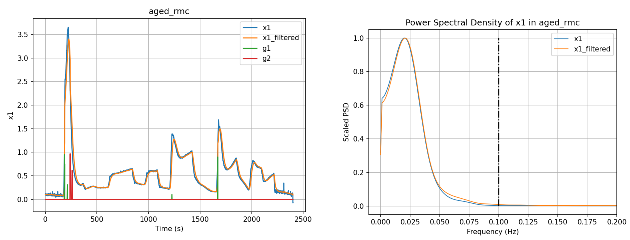

Figure 43: Filtered results for x_1

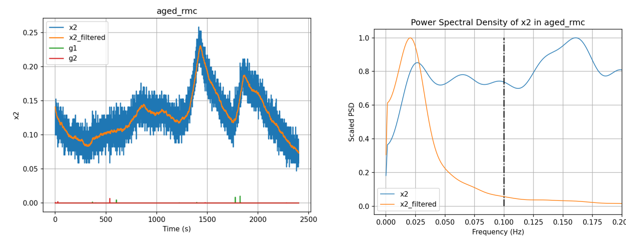

Figure 44: Filtered results for x_2

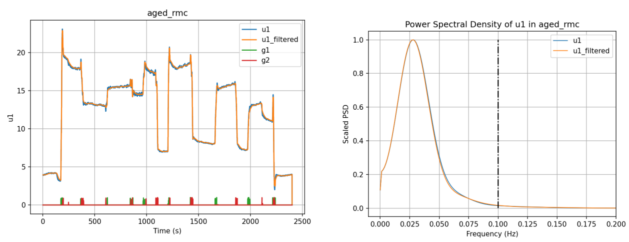

Figure 45: Filtered results for u_1

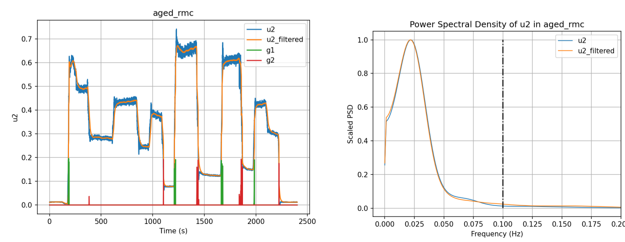

Figure 46: Filtered results for u_2

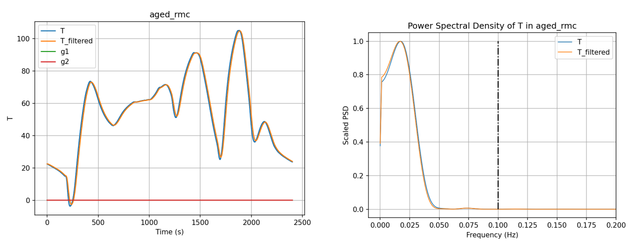

Figure 47: Filtered results for T

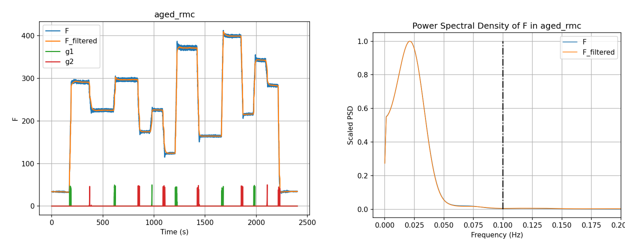

Figure 48: Filtered results for F

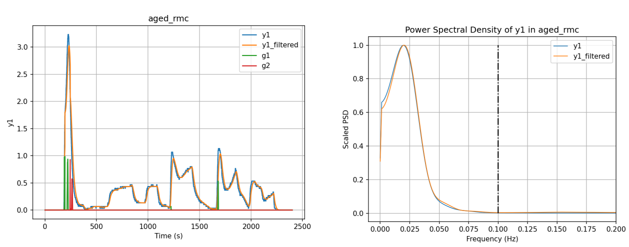

Figure 49: Filtered results for y_1

9 Test Data Signal Ranges

The ranges of the measurement signals are tabulated bellow:

Variable	Units			
Degreend Data		Cold FTP	Hot FTP	RMC
T	$+200\,^0C$	[60.96, -174.0]	[67.38, -64.7]	[148.76, 43.76]
F	g/s	[439.04, 0.0]	[431.14, 0.0]	[403.84, 32.65]
x_1	mol/m^3	[17.24, -2.97]	[3.55, -0.57]	[2.23, 0.14]
x_2	mol/m^3	[0.06, 0.0]	[0.17, -0.01]	[0.27, 0.06]
u_1	mol/m^3	[20.16, 0.0]	[16.46, 0.0]	[24.26, 0.0]
u_2	mol/m^3	[1.4, 0.0]	[1.0, 0.0]	[0.71, 0.01]
y_1	mol/m^3	[0.07, 0.0]	[0.82, 0.0]	[2.11, 0.0]
Aged Data		Cold FTP	Hot FTP	RMC
T	$+200\,^0C$	[67.03, -172.85]	[72.5, -56.35]	[154.38, 48.18]
F	g/s	[416.85, 0.0]	[418.24, 0.0]	[409.59, 32.63]
x_1	mol/m^3	[25.03, 0.01]	[4.9, -0.42]	[3.24, 0.08]
x_2	mol/m^3	[0.04, 0.0]	[0.6, -0.02]	[0.23, 0.08]
u_1	mol/m^3	[18.58, 0.0]	[16.26, 0.0]	[22.89, 0.0]
u_2	mol/m^3	[1.31, 0.0]	[1.0, 0.0]	[0.67, 0.01]
y_1	mol/m^3	[0.09, 0.0]	[1.06, 0.0]	[2.86, 0.0]

Table 2: Test cell data ranges

The variable names for each state in the data-set are tabulated bellow for reference.

Variable Name	Units	Description	Variable from Model
LOG_TM	sec	Time	t
ENG_SPD	RPM	Engine Speed	
NET_TORQ	lb_ft	Engine Torque	
TEST_CELL_T	deg_f	Test Cell Temperature	
TUR_OT_T	deg_f	Turbine Outlet Gas Temperature	
EXHAUST_FLOW	kg/min	Exhaust Flow Rate	F
V_AIM_TRC_DOC_IN	Deg_C	DOC In Gas Temperature	
V_AIM_TRC_DOC_OUT	Deg_C	DOC Out Gas Temperature	
V_AIM_TRC_DPF_OUT	Deg_C	DPF Out Gas Temperature	
V_AIM_TRC_SCR_OUT	Deg_C	SCR/ASC Out Gas Temperature	
V_SCR_ANR_FDBK	None	ANR (??)	
V_UIM_FLM_ESTUREAINJRATE	ml/sec	DEF (Urea Sol.) Dosing Rate	u_2
ENG_CW_NOX_FTIR_COR_U2	PPM	Engine-Out NOx	u_1
EXH_CW_NOX_COR_U1	PPM	Tailpipe NOx	x_1
EXH_CW_AMMONIA_MEA	ppm	Tailpipe NH3	x_2
EXH_CW_N2O_MEA	ppm	Tailpipe N2O	
ENG_CW_NO2_COR_U2	PPM	DPF Outlet NO2	
EONOX_COMP_VALUE	ppm	Engine-out NOx (with Corss-sensitivity)	y_3
V_SCM_PPM_SCR_OUT_NOX	ppm	SCR-out NOx (with cross-sensitivity)	y_1

Table 3: Test Cell Data Variables

Part II

Truck Data Signal Processing

10 Power Spectral Density of Individual Truck Signals

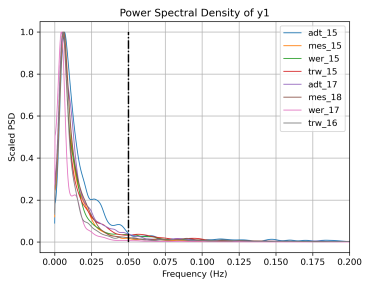

Figure 50: PSD of $[NO_x]^{out}$

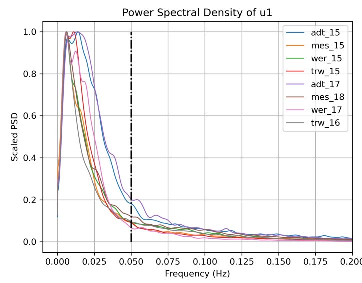

Figure 51: PSD of $[NO_x]^{in}$

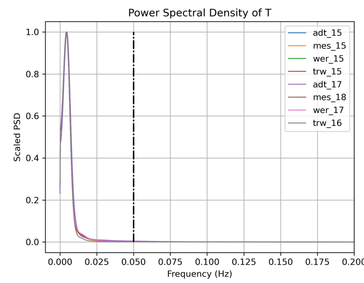

Figure 52: PSD of temperature

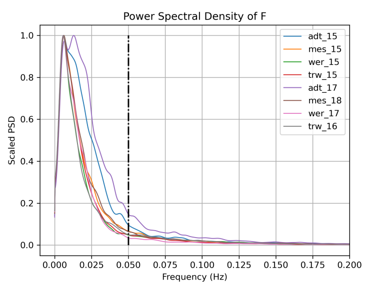

Figure 53: PSD of mass flow rate

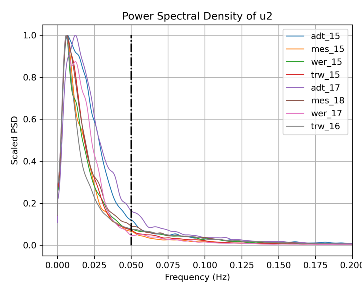

Figure 54: PSD of $[NO_x]^{out}$

10.1 Forgetting factor for RLS

The PSD plots show that much of the energy for the signals in within the frequency range of $0\,Hz$ to $0.05\,Hz$. Thus, the forgetting factor for the RLS algorithm is chosen such that the equivalent time-constant is greater than $1/0.05 = 20\,s$. We have,

$$T_f = \frac{t_d}{1-\lambda}$$
$$T_f = 20 \quad \text{and} \quad t_d = 1$$
$$\implies \lambda = 1 - \frac{t_d}{T_f} = 1 - \frac{1}{20} = 0.95$$

10.2 h and ν parameters for individual signals

signal	h	ν
y_1	5	2.5
u_1	40	20
u_2	2	1
T	40	30
F	400	200

Table 4: Values of threshold (h) and bias (ν) for individual truck data signals

11 AD Transport 2015

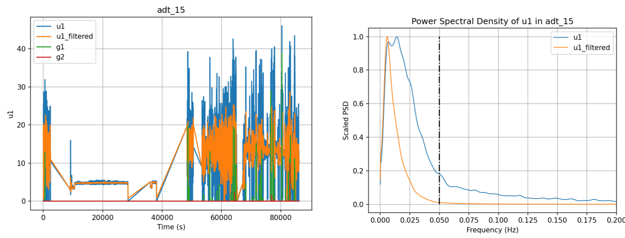

Figure 55: Filtered results for u_1

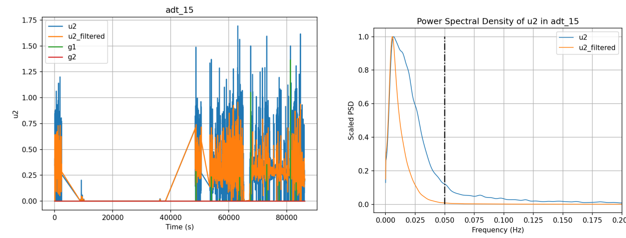

Figure 56: Filtered results for u_2

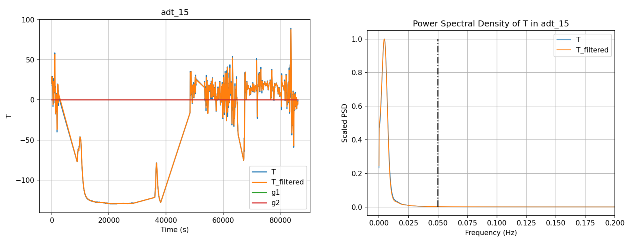

Figure 57: Filtered results for T

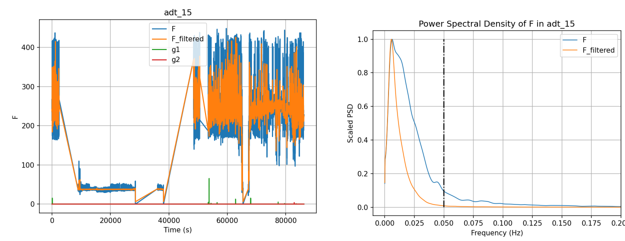

Figure 58: Filtered results for F

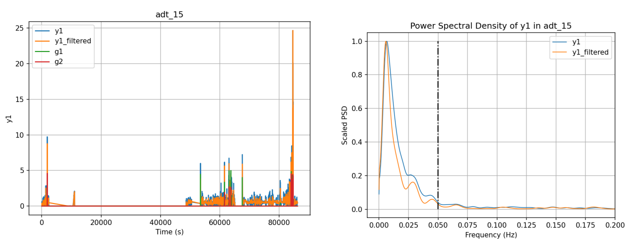

Figure 59: Filtered results for y_1

12 AD Transport 2017

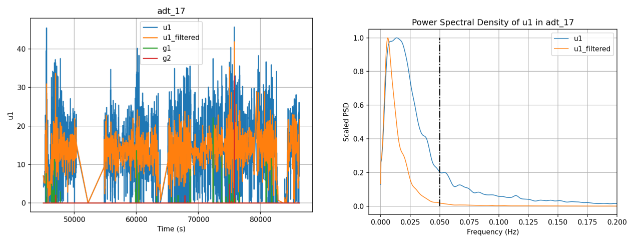

Figure 60: Filtered results for u_1

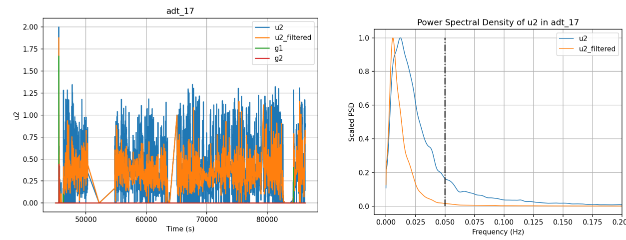

Figure 61: Filtered results for u_2

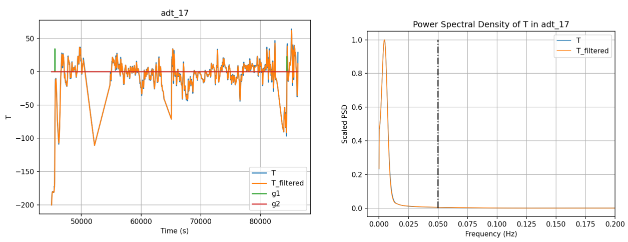

Figure 62: Filtered results for T

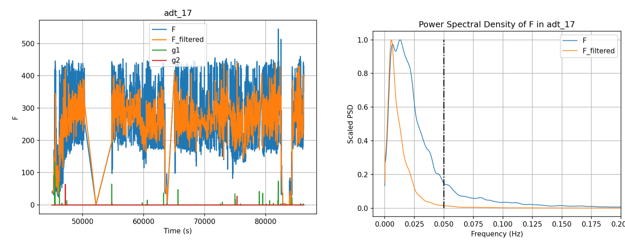

Figure 63: Filtered results for F

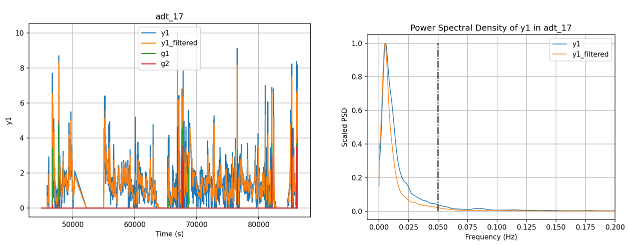

Figure 64: Filtered results for y_1

13 Mesilla Valley 2015

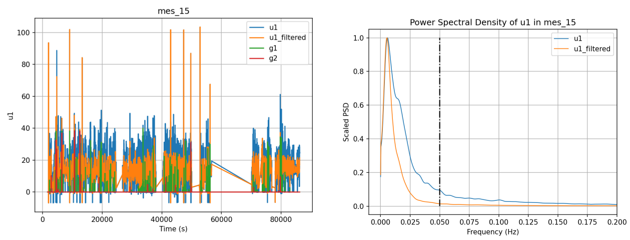

Figure 65: Filtered results for u_1

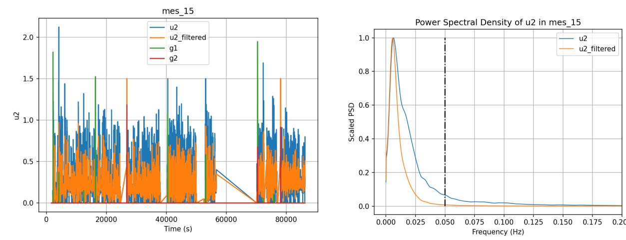

Figure 66: Filtered results for u_2

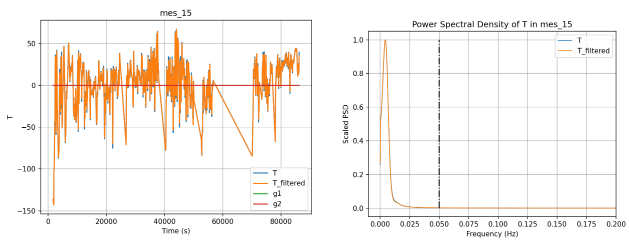

Figure 67: Filtered results for T

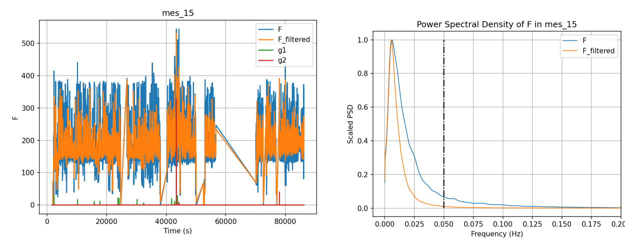

Figure 68: Filtered results for F

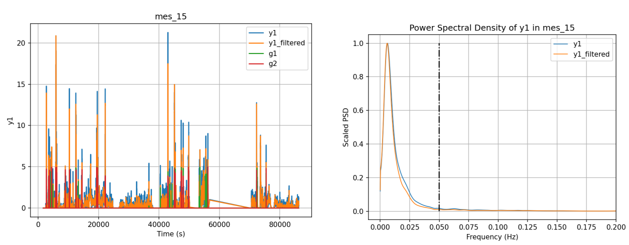

Figure 69: Filtered results for y_1

14 Mesilla Valley 2018

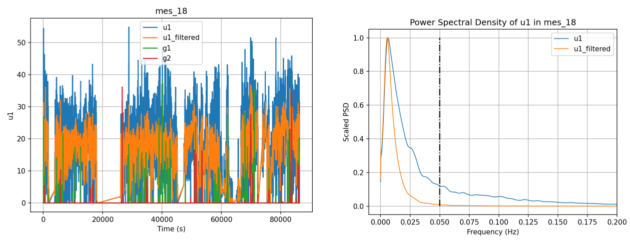

Figure 70: Filtered results for u_1

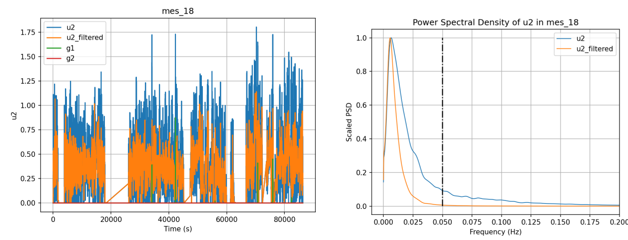

Figure 71: Filtered results for u_2

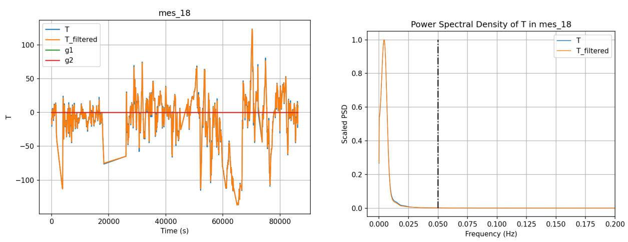

Figure 72: Filtered results for T

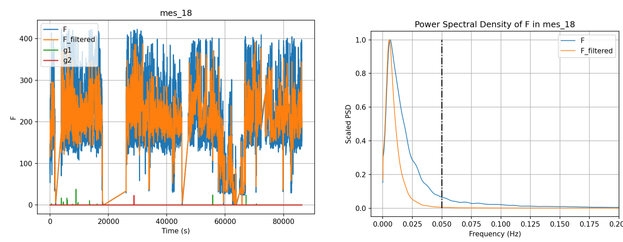

Figure 73: Filtered results for F

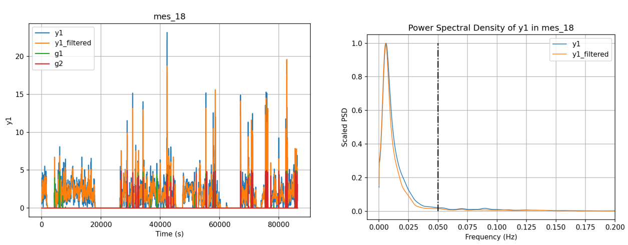

Figure 74: Filtered results for y_1

15 Transwest 2015

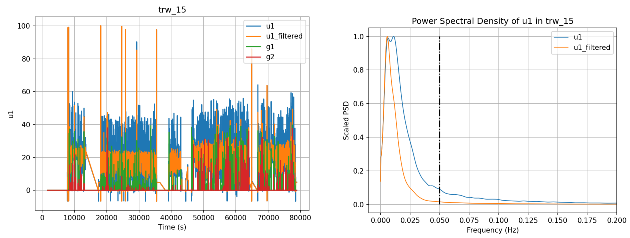

Figure 75: Filtered results for u_1

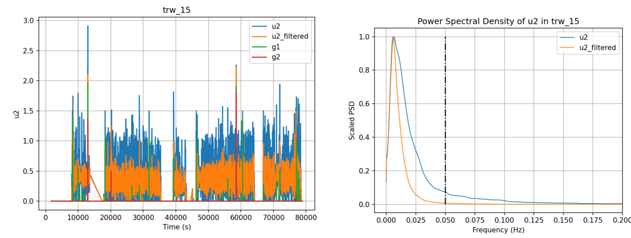

Figure 76: Filtered results for u_2

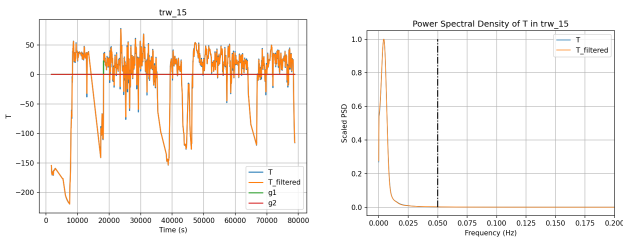

Figure 77: Filtered results for T

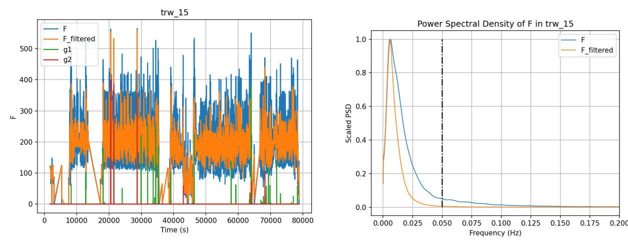

Figure 78: Filtered results for F

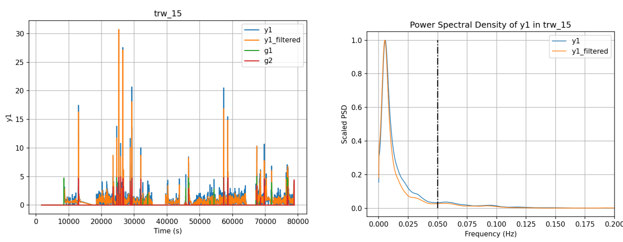

Figure 79: Filtered results for y_1

16 Transwest 2016

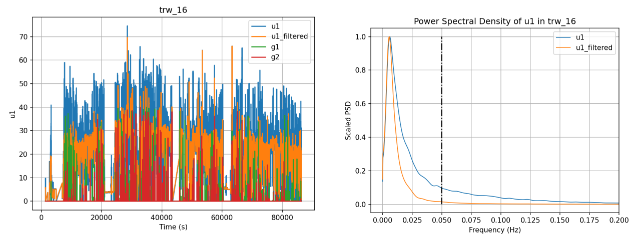

Figure 80: Filtered results for u_1

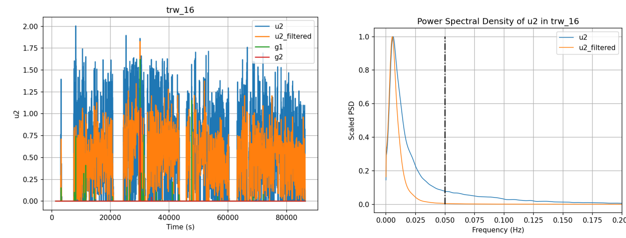

Figure 81: Filtered results for u_2

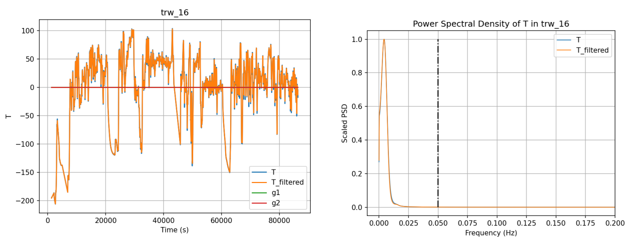

Figure 82: Filtered results for T

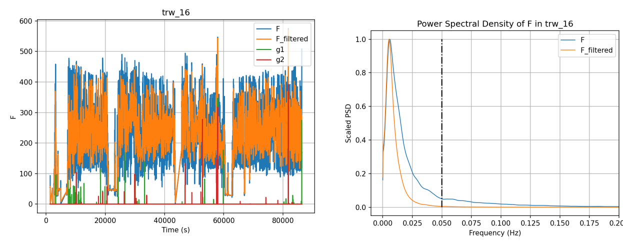

Figure 83: Filtered results for F

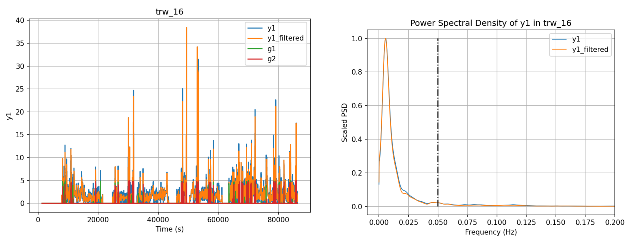

Figure 84: Filtered results for y_1

17 Werner 2015

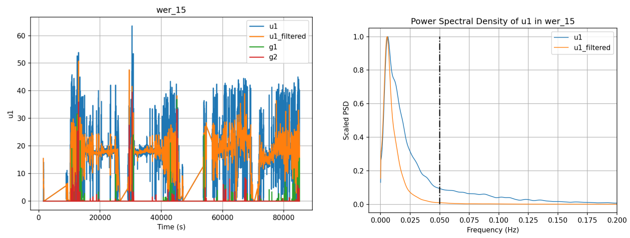

Figure 85: Filtered results for u_1

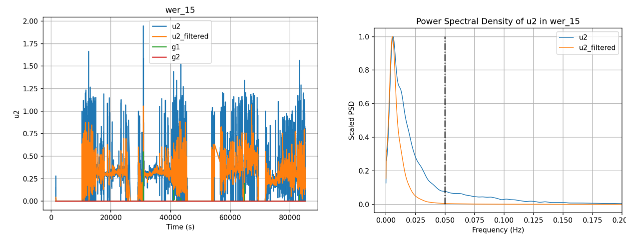

Figure 86: Filtered results for u_2

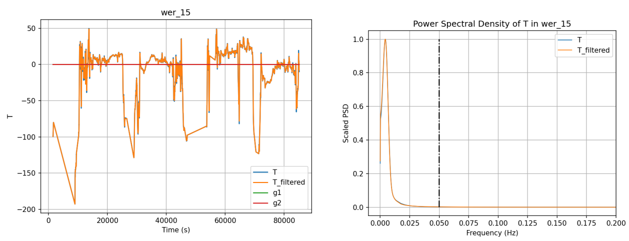

Figure 87: Filtered results for T

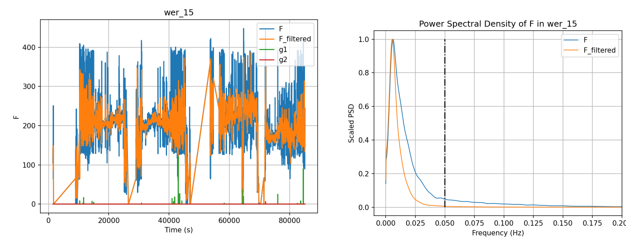

Figure 88: Filtered results for F

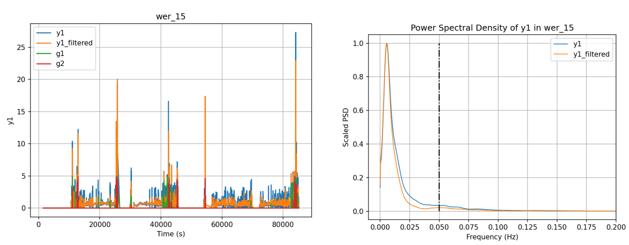

Figure 89: Filtered results for y_1

18 Werner 2017

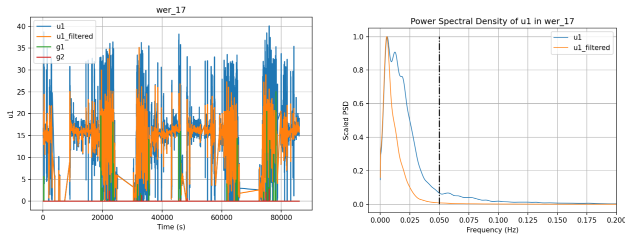

Figure 90: Filtered results for u_1

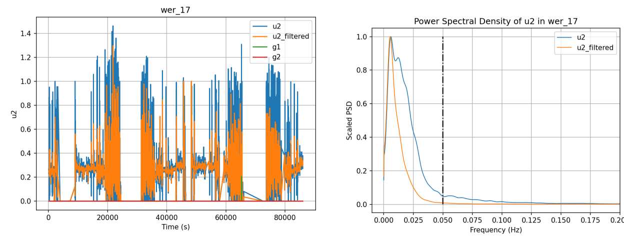

Figure 91: Filtered results for u_2

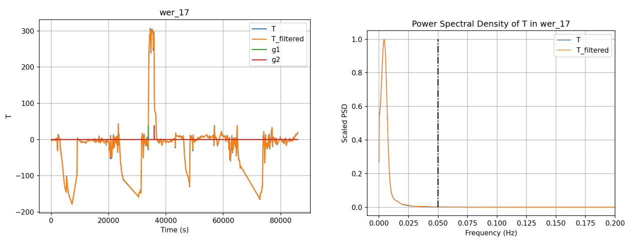

Figure 92: Filtered results for T

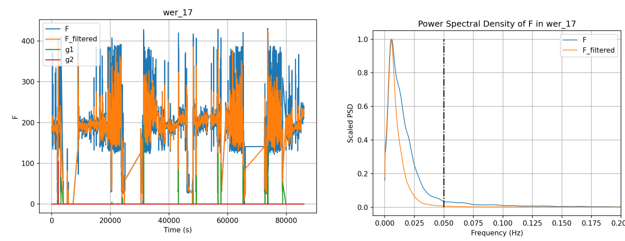

Figure 93: Filtered results for F

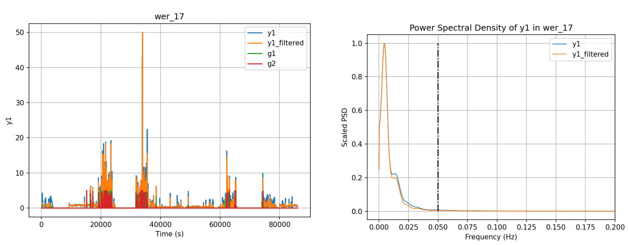

Figure 94: Filtered results for y_1

19 Truck Data Signal Ranges

The ranges of the measurement signals are tabulated bellow:

Variable	Units				
Degreend Data		adt_15	mes_15	wer_15	trw_15
T	$+200\,^0C$	[135.71, -79.68]	[116.91, -93.05]	[98.7, -142.85]	[125.21, -169.55]
F	g/s	[425.94, 15.82]	[529.08, 0.0]	[386.51, 0.0]	[559.86, 0.0]
u_1	mol/m^3	[27.34, 0.0]	[103.47, -6.87]	[49.99, 0.0]	[100.22, -6.73]
u_2	mol/m^3	[0.89, 0.0]	[1.5, 0.0]	[1.07, 0.0]	[2.18, 0.0]
y_1	mol/m^3	[24.66, 0.0]	[20.86, 0.0]	[22.51, 0.0]	[30.73, 0.0]
Aged Data		adt_17	mes_18	wer_20	trw_16
T	$+200\,^0C$	[111.04, -150.35]	[173.4, -86.67]	[353.84, -127.68]	[152.48, -155.45]
F	g/s	[440.63, 0.0]	[391.06, 0.0]	[425.77, 0.0]	[569.7, 0.0]
u_1	mol/m^3	[41.76, 0.0]	[36.89, 0.0]	[35.16, 0.0]	[68.93, 0.0]
u_2	mol/m^3	[1.88, 0.0]	[1.07, 0.0]	[1.28, 0.0]	[1.66, 0.0]
y_1	mol/m^3	[9.14, 0.0]	[19.6, 0.0]	[49.99, 0.0]	[38.39, 0.0]

Table 5: Truck data ranges

The variable names for each state in the data-set are tabulated bellow for reference.

Model Variable	Truck Data Variable	Units
t	tod	s
T	pSCRBedTemp	^{0}C
F	pExhMF	g/s
x_1		ppm
x_2		ppm
u_1		ppm
u_2	pUreaDosing	ml/sec
y_1	pNOxOutppm	ppm
u_1	pNOxInppm	ppm

Table 6: Truck data variables

20 Conclusion

The tuning parameters for filtering individual truck and test data signals are obtained. The validity of filtered signals is verified by using the power spectral density plots by making sure the curve remains similar within the frequency range of interest which is different for test and truck cases. The results are tabulated bellow:

Table 7: Parameters for test data

signal	h	ν
x_1	1	0.5
x_2	0.04	0.04
u_1	1	0.5
u_2	0.2	0.1
T	40	30
F	50	20
y_1	1	0.5

$$\lambda = 0.98$$

Frequency range : $[0, 0.1]$ Hz

Table 8: Parameters for truck data

signal	h	ν
y_1	5	2.5
u_1	40	20
u_2	2	1
T	40	30
F	400	200

$$\lambda = 0.95$$

Frequency range : $[0, 0.05]$ Hz

The results of this work are used for parameter estimation of the discrete nonlinear model using windowed-least squares. The nonlinear model considers the residence time of the reactants under plug-flow reactor assumptions.

References

[1] Maurice S Bartlett. "Periodogram analysis and continuous spectra". In: *Biometrika* 37.1/2 (1950), pp. 1–16.

[2] Fredrik Gustafsson and Fredrik Gustafsson. *Adaptive filtering and change detection.* Vol. 1. Wiley New York, 2000.

[3] Peter Welch. "The use of fast Fourier transform for the estimation of power spectra: a method based on time averaging over short, modified periodograms". In: *IEEE Transactions on audio and electroacoustics* 15.2 (1967), pp. 70–73.

[4] Bin Yao. "System Identification and Parameter Estimation". In: *Lecture Notes* ().